T9 Algoritmo backprop

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1 Introducción

Objetivo: introducir el algoritmo backprop para entrenar perceptrones multicapa con ejemplos de uso

Backprop: primero veremos el algoritmo backprop genérico para hallar el gradiente de un modelo

Ejemplo con una red para regresión: cálculo del gradiente y actualización de parámetros

- A partir de un dato
- Generalización de dato a batch

XOR: cálculo del gradiente y actualización de parámetros

- Ejemplo: red con una capa oculta de dos ReLUs y salida softmax
- Ejercicio: análogo al ejemplo, pero con salida sigmoide en lugar de softmax

2 Algoritmo backprop

Algoritmo backprop (backpropagation):

- Calcula el gradiente de la pérdida con respecto a los parámetros de cada capa
- El gradiente calculado se pasa a un algoritmo de optimización
- Origen: 1982; Paul Werbos lo aplica a MLPs de la manera que ha pasado a ser estándar
- Popularización: Rumelhart, Hinton y Williams lo popularizan en un artículo de Nature de 1986

Entrada: MLP de $L-1$ capas, cada una con su $\boldsymbol{\theta}^{(\ell)}$ ($\emptyset$ si no tiene), y pérdida respecto a $(\boldsymbol{x}, \boldsymbol{y})$ en "capa" L

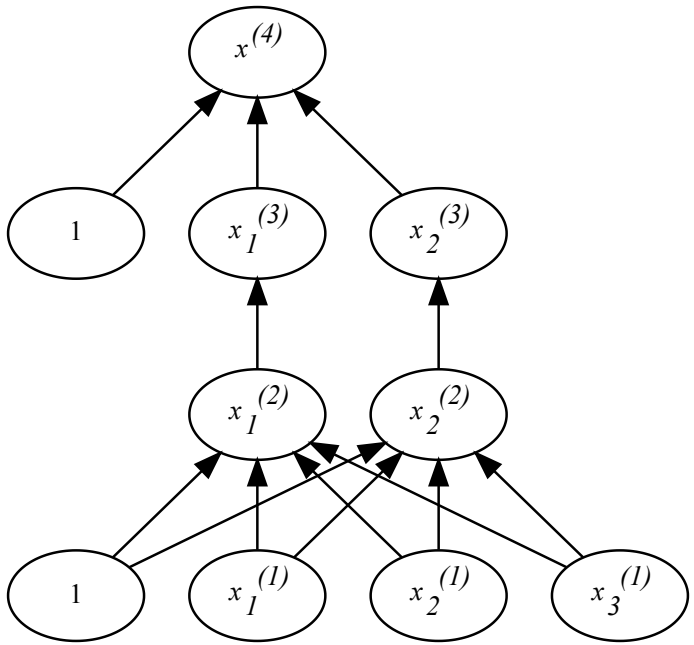
$$\boldsymbol{x}^{(1)} = \boldsymbol{x} \;\rightarrow\; \boldsymbol{x}^{(2)} = \boldsymbol{f}^{(1)}(\boldsymbol{x}^{(1)}, \boldsymbol{\theta}^{(1)}) \;\overset{\cdots}{\rightarrow}\; \boldsymbol{x}^{(L)} = \boldsymbol{f}^{(L-1)}(\boldsymbol{x}^{(L-1)}, \boldsymbol{\theta}^{(L-1)}) \;\rightarrow\; x^{(L+1)} = \boldsymbol{f}^{(L)}(\boldsymbol{x}^{(L)}, \boldsymbol{y}) = \mathcal{L}$$

Método: bucle que ejecuta dos pasos en cada iteración

- **Forward:** recorre el modelo hacia adelante para hallar la salida de cada capa a partir de una entrada $\boldsymbol{x}$
  - $\boldsymbol{x}^{(1)} = \boldsymbol{x}$
  - **para** $\ell = 1 : L :$ $\quad \boldsymbol{x}^{(\ell+1)} = \boldsymbol{f}^{(\ell)}(\boldsymbol{x}^{(\ell)}, \boldsymbol{\theta}^{(\ell)})$
- **Backward:** recorre el modelo hacia atrás para hallar el gradiente de la pérdida respecto a los parámetros
  - $u^{(L+1)} = 1$
  - **para** $\ell = L : 1 :$ $\quad \boldsymbol{g}^{(\ell)} = \boldsymbol{u}^{(\ell+1)t} \dfrac{\partial \boldsymbol{x}^{(\ell+1)}}{\partial \boldsymbol{\theta}^{(\ell)}} \qquad \boldsymbol{u}^{(\ell)t} = \boldsymbol{u}^{(\ell+1)t} \dfrac{\partial \boldsymbol{x}^{(\ell+1)}}{\partial \boldsymbol{x}^{(\ell)}}$

3 Ejemplo de regresión con un dato

In [ ]:
```
import graphviz; graphviz.Source('''digraph {rankdir=BT; splines=false; node[shape=oval fontsize=11]
{rank=same x10[label=1] x11[label=<<i>x<sub>1</sub><sup>(1)</sup></i>>]
  x12[label=<<i>x<sub>2</sub><sup>(1)</sup></i>>] x13[label=<<i>x<sub>3</sub><sup>(1)</sup></i>>]}
{rank=same x21[label=<<i>x<sub>1</sub><sup>(2)</sup></i>>] x22[label=<<i>x<sub>2</sub><sup>(2)</sup></i>>]}
{rank=same x30[label=1] x31[label=<<i>x<sub>1</sub><sup>(3)</sup></i>>]
  x32[label=<<i>x<sub>2</sub><sup>(3)</sup></i>>]} y[label=<<i>x<sup>(4)</sup></i>>]
{x10 x11 x12 x13} -> {x21 x22} x21 -> x31 x22 -> x32 {x30 x31 x32} -> y edge[style=invis]
x10->x11->x12->x13 x21->x22 x30->x31->x32}''').render(filename='3_Ejemplo_un_dato', format='svg');
```



$$\boldsymbol{x}^{(1)} \in \mathbb{R}^3 \;\rightarrow\; \boldsymbol{x}^{(2)} = \mathbf{W}^{(1)}\boldsymbol{x}^{(1)} + \boldsymbol{b}^{(1)} \in \mathbb{R}^2 \;\rightarrow\; \boldsymbol{x}^{(3)} = \boldsymbol{\sigma}(\boldsymbol{x}^{(2)}) \in \mathbb{R}^2 \;\rightarrow\; x^{(4)} = \mathbf{W}^{(3)}\boldsymbol{x}^{(3)} + b^{(3)} \in \mathbb{R}$$

Objetivo: dado $(\boldsymbol{x}, y) = (\mathbf{1}, 1)$, minimizar la pérdida cuadrática mediante una iteración de GD con $\eta = 0.1$

Parámetros actuales: supongamos que $\boldsymbol{\theta} = \mathbf{1}$, o, siguiendo el formato del modelo,

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \qquad \boldsymbol{b}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \mathbf{W}^{(3)} = (1, 1) \qquad b^{(3)} = 1$$

Pérdida: para hallar el gradiente de la pérdida cuadrática, consideramos la pérdida como una "capa" adicional

$$\boldsymbol{x}^{(1)} = \boldsymbol{x} \;\rightarrow\; \boldsymbol{x}^{(2)} \;\rightarrow\; \boldsymbol{x}^{(3)} \;\rightarrow\; x^{(4)} = \hat{y} \;\rightarrow\; x^{(5)} = \mathcal{L} = (y - \hat{y})^2$$

Forward: recorremos el modelo hacia adelante para hallar la salida de cada capa a partir de la entrada

$$\boldsymbol{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\boldsymbol{x}^{(2)} = \mathbf{W}^{(1)}\boldsymbol{x}^{(1)} + \boldsymbol{b}^{(1)} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$$\boldsymbol{x}^{(3)} = \boldsymbol{\sigma}(\boldsymbol{x}^{(2)}) = \boldsymbol{\sigma}\left(\begin{pmatrix} 4 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} \sigma(4) \\ \sigma(4) \end{pmatrix} = \begin{pmatrix} 0.982 \\ 0.982 \end{pmatrix}$$

$$\hat{y} = x^{(4)} = \mathbf{W}^{(3)}\boldsymbol{x}^{(3)} + b^{(3)} = (1, 1) \begin{pmatrix} 0.982 \\ 0.982 \end{pmatrix} + 1 = 2.9640$$

$$\mathcal{L} = x^{(5)} = (y - \hat{y})^2 = (1 - 2.9640)^2 = 3.8574$$

Backward: recorremos el modelo hacia atrás para hallar el gradiente de la pérdida respecto a los parámetros

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - y) = 3.9280$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{W}^{(3)}} = \boldsymbol{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \hat{y}} = \begin{pmatrix} 3.8574 \\ 3.8574 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot 1 = 3.9280$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \mathbf{W}^{(3)} = (3.9280, 3.9280)$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \frac{\partial \boldsymbol{x}^{(3)}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \odot \boldsymbol{\sigma}'(\boldsymbol{x}^{(2)t}) = (0.0694, 0.0694)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = \begin{pmatrix} 0.0694 & 0.0694 \\ 0.0694 & 0.0694 \\ 0.0694 & 0.0694 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \odot \mathbf{1}^t = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = (0.0694, 0.0694)$$

Actualización de parámetros: lo hacemos capa a capa, respetando el formato del modelo

$$\begin{aligned}\mathbf{W}^{(3)} &= \mathbf{W}^{(3)} - \eta\left(\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}}\right)^t \\ &= (1,1) - 0.1\,(3.8574, 3.8574) = (0.6143, 0.6143)\end{aligned}$$

$$\begin{aligned}b^{(3)} &= b^{(3)} - \eta\left(\frac{\partial \mathcal{L}}{\partial b^{(3)}}\right)^t \\ &= 1 - 0.1 \cdot 3.9280 = 0.6072\end{aligned}$$

$$\begin{aligned}\mathbf{W}^{(1)} &= \mathbf{W}^{(1)} - \eta\left(\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}}\right)^t \\ &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0694 & 0.0694 & 0.0694 \\ 0.0694 & 0.0694 & 0.0694 \end{pmatrix} = \begin{pmatrix} 0.9931 & 0.9931 & 0.9931 \\ 0.9931 & 0.9931 & 0.9931 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\boldsymbol{b}^{(1)} &= \boldsymbol{b}^{(1)} - \eta\left(\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}}\right)^t \\ &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0694 \\ 0.0694 \end{pmatrix} = \begin{pmatrix} 0.9931 \\ 0.9931 \end{pmatrix}\end{aligned}$$

In [ ]:

```python
import numpy as np; np.set_printoptions(precision=4); import torch; from torch import nn
class NeuralNetwork(nn.Module):
    def __init__(self):
        super().__init__()
        self.x2 = nn.Linear(3, 2); nn.init.ones_(self.x2.weight); nn.init.ones_(self.x2.bias)
        self.x3 = nn.Sigmoid()
        self.x4 = nn.Linear(2, 1); nn.init.ones_(self.x4.weight); nn.init.ones_(self.x4.bias)
    def forward(self, x):
        return self.x4(self.x3(self.x2(x)))
model = NeuralNetwork(); print(model)
x = torch.ones(1, 3); print(f'x = {x.detach()}')
y = torch.ones(1, 1); print(f'y = {y.detach()}')
x2 = model.x2(x); print(f'x2 = {x2.detach()}')
x3 = model.x3(x2); print(f'x3 = {x3.detach()}')
x4 = model.x4(x3); print(f'x4 = {x4.detach()}')
loss = nn.MSELoss()(x4, y); print(f'loss = {loss:.4f}')
model.train(); loss.backward()
torch.optim.SGD(model.parameters(), lr=1e-1).step()
print(f'W3 = {model.x4.weight.detach()}')
print(f'b3 = {model.x4.bias.detach()}')
print(f'W1 = {model.x2.weight.detach()}')
print(f'b1 = {model.x2.bias.detach()}')
```

```
NeuralNetwork(
  (x2): Linear(in_features=3, out_features=2, bias=True)
  (x3): Sigmoid()
  (x4): Linear(in_features=2, out_features=1, bias=True)
)
x = tensor([[1., 1., 1.]])
y = tensor([[1.]])
x2 = tensor([[4., 4.]])
x3 = tensor([[0.9820, 0.9820]])
x4 = tensor([[2.9640]])
loss = 3.8574
W3 = tensor([[0.6143, 0.6143]])
b3 = tensor([0.6072])
W1 = tensor([[0.9931, 0.9931, 0.9931],
        [0.9931, 0.9931, 0.9931]])
b1 = tensor([0.9931, 0.9931])
```

4 Ejemplo de regresión con un batch

Objetivo: entrenar el modelo anterior con $N = 2$ datos, $\mathbf{X} = \begin{pmatrix} \boldsymbol{x}_1^t \\ \boldsymbol{x}_2^t \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$ y $\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Forward: la generalización de dato a batch es sencilla

$$\mathbf{X}^{(1)} = \mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$$

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)}\mathbf{W}^{(1)t} + \mathbf{1}\boldsymbol{b}^{(1)t} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1, 1) = \begin{pmatrix} 4 & 4 \\ 7 & 7 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \boldsymbol{\sigma}(\mathbf{X}^{(2)}) = \boldsymbol{\sigma}\left(\begin{pmatrix} 4 & 4 \\ 7 & 7 \end{pmatrix}\right) = \begin{pmatrix} \sigma(4) & \sigma(4) \\ \sigma(7) & \sigma(7) \end{pmatrix} = \begin{pmatrix} 0.9820 & 0.9820 \\ 0.9991 & 0.9991 \end{pmatrix}$$

$$\hat{\boldsymbol{y}} = \mathbf{X}^{(4)} = \mathbf{X}^{(3)}\mathbf{W}^{(3)t} + \mathbf{1}b^{(3)t} = \begin{pmatrix} 0.9820 & 0.9820 \\ 0.9991 & 0.9991 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} 1 = \begin{pmatrix} 2.9640 \\ 2.9982 \end{pmatrix}$$

$$\boldsymbol{y} - \hat{\boldsymbol{y}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2.9640 \\ 2.9982 \end{pmatrix} = \begin{pmatrix} -1.9640 \\ -0.9982 \end{pmatrix}$$

$$\mathcal{L} = x^{(5)} = \frac{1}{N}\sum_n (y_n - \hat{y}_n)^2 = \frac{1}{N}(\boldsymbol{y} - \hat{\boldsymbol{y}})^t(\boldsymbol{y} - \hat{\boldsymbol{y}}) = 2.4268$$

Matrices de pesos y vectores de sesgos transpuestos: mayor eficiencia en forward (e inferencia)

Backward: generalización de dato a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - y) \qquad \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} = \frac{2}{N}(\hat{\boldsymbol{y}} - \boldsymbol{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \mathbf{W}^{(3)}} = \boldsymbol{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \hat{y}} \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b^{(3)}} \qquad \frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \mathbf{W}^{(3)} \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \frac{\partial \boldsymbol{x}^{(3)}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \odot \boldsymbol{\sigma}'(\boldsymbol{x}^{(2)t}) \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \odot \mathbf{1}^t \qquad \frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} = \frac{2}{N}(\hat{\boldsymbol{y}} - \boldsymbol{y})^t = (1.9640, 0.9982)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t}\left(\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}}\right)^t = \begin{pmatrix} 0.9820 & 0.9991 \\ 0.9820 & 0.9991 \end{pmatrix}\begin{pmatrix} 1.9640 \\ 0.9982 \end{pmatrix} = \begin{pmatrix} 2.9259 \\ 2.9259 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}}\right)^t = (1,1)\begin{pmatrix} 1.9640 \\ 0.9982 \end{pmatrix} = 2.9622$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t}\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}(1.9640, 0.9982) = \begin{pmatrix} 1.9640 & 0.9982 \\ 1.9640 & 0.9982 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \boldsymbol{\sigma}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 1.9640 & 0.9982 \\ 1.9640 & 0.9982 \end{pmatrix} \odot \begin{pmatrix} \sigma'(4) & \sigma'(7) \\ \sigma'(4) & \sigma'(7) \end{pmatrix} = \begin{pmatrix} 0.0347 & 0.0009 \\ 0.0347 & 0.0009 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = \begin{pmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 0.0347 & 0.0347 \\ 0.0009 & 0.0009 \end{pmatrix} = \begin{pmatrix} 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = (1,1)\begin{pmatrix} 0.0347 & 0.0347 \\ 0.0009 & 0.0009 \end{pmatrix} = (0.0356, 0.0356)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 0.1 \begin{pmatrix} 2.9259 \\ 2.9259 \end{pmatrix} = \begin{pmatrix} 0.7074 \\ 0.7074 \end{pmatrix}$$

$$b^{(3)t} = b^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial b^{(3)}} = 1 - 0.1 \cdot 2.9622 = 0.7038$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \\ 0.0365 & 0.0365 \end{pmatrix} = \begin{pmatrix} 0.9964 & 0.9964 \\ 0.9964 & 0.9964 \\ 0.9964 & 0.9964 \end{pmatrix}$$

$$\boldsymbol{b}^{(1)t} = \boldsymbol{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = (1, 1) - 0.1(0.0356, 0.0356) = (0.9964, 0.9964)$$

In [ ]:

```python
import numpy as np; np.set_printoptions(precision=4); import torch; from torch import nn
class NeuralNetwork(nn.Module):
    def __init__(self):
        super().__init__()
        self.x2 = nn.Linear(3, 2); nn.init.ones_(self.x2.weight); nn.init.ones_(self.x2.bias)
        self.x3 = nn.Sigmoid()
        self.x4 = nn.Linear(2, 1); nn.init.ones_(self.x4.weight); nn.init.ones_(self.x4.bias)
    def forward(self, x):
        return self.x4(self.x3(self.x2(x)))
model = NeuralNetwork(); print(model)
X = torch.tensor([[1., 1., 1.], [2., 2., 2.]]); print(f'X = {str(X.detach()).replace('\n','')}')
y = torch.tensor([[1.], [2.]]); print(f'y = {str(y.detach()).replace('\n','')}')
X2 = model.x2(X); print(f'X2 = {str(X2.detach()).replace('\n','')}')
X3 = model.x3(X2); print(f'X3 = {str(X3.detach()).replace('\n','')}')
X4 = model.x4(X3); print(f'X4 = {str(X4.detach()).replace('\n','')}')
loss = nn.MSELoss()(X4, y); print(f'loss = {loss:.4f}')
model.train(); loss.backward()
torch.optim.SGD(model.parameters(), lr=1e-1).step()
print(f'W3 = {model.x4.weight.detach()}')
print(f'b3 = {model.x4.bias.detach()}')
print(f'W1 = {model.x2.weight.detach()}')
print(f'b1 = {model.x2.bias.detach()}')
```

```
NeuralNetwork(
  (x2): Linear(in_features=3, out_features=2, bias=True)
  (x3): Sigmoid()
  (x4): Linear(in_features=2, out_features=1, bias=True)
)
X = tensor([[1., 1., 1.],        [2., 2., 2.]])
y = tensor([[1.],        [2.]])
X2 = tensor([[4., 4.],        [7., 7.]])
X3 = tensor([[0.9820, 0.9820],        [0.9991, 0.9991]])
X4 = tensor([[2.9640],        [2.9982]])
loss = 2.4269
W3 = tensor([[0.7074, 0.7074]])
b3 = tensor([0.7038])
W1 = tensor([[0.9963, 0.9963, 0.9963],
        [0.9963, 0.9963, 0.9963]])
b1 = tensor([0.9964, 0.9964])
```

5 Ejemplo para XOR con salida softmax

Modelo: capa oculta con dos ReLUs y capa de salida con softmax para dos clases

$$\boldsymbol{x}^{(1)} \in \{0,1\}^2$$

$$\boldsymbol{x}^{(2)} = \mathbf{W}^{(1)}\boldsymbol{x}^{(1)} + \boldsymbol{b}^{(1)} \in \mathbb{R}^2$$

$$\boldsymbol{x}^{(3)} = \mathrm{ReLU}(\boldsymbol{x}^{(2)}) \in \mathbb{R}^2$$

$$\boldsymbol{x}^{(4)} = \mathbf{W}^{(3)}\boldsymbol{x}^{(3)} + \boldsymbol{b}^{(3)} \in \mathbb{R}^2$$

$$\boldsymbol{x}^{(5)} = \hat{\boldsymbol{y}} = \mathcal{S}(\boldsymbol{x}^{(4)}) \in [0,1]^2$$

$$x^{(6)} = \mathcal{L} = \mathbb{H}(\boldsymbol{y}, \hat{\boldsymbol{y}}) = -\boldsymbol{y}^t \log(\hat{\boldsymbol{y}})$$

Parámetros actuales:

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} -1 \\ 0.5 \end{pmatrix} \quad \mathbf{W}^{(3)} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \boldsymbol{b}^{(3)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Objetivo: backprop con $\mathbf{X} = \begin{pmatrix} \boldsymbol{x}_1^t \\ \boldsymbol{x}_2^t \\ \boldsymbol{x}_3^t \\ \boldsymbol{x}_4^t \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$ y $\mathbf{Y} = \begin{pmatrix} \boldsymbol{y}_1^t \\ \boldsymbol{y}_2^t \\ \boldsymbol{y}_3^t \\ \boldsymbol{y}_4^t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$

Forward:

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)}\mathbf{W}^{(1)t} + \mathbf{1}\boldsymbol{b}^{(1)t} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (-1, 0.5) = \begin{pmatrix} -1 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \mathrm{ReLU}(\mathbf{X}^{(2)}) = \begin{pmatrix} 0 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix}$$

$$\mathbf{X}^{(4)} = \mathbf{X}^{(3)}\mathbf{W}^{(3)t} + \mathbf{1}\boldsymbol{b}^{(3)t} = \begin{pmatrix} 0 & 0.5 \\ 0 & 1.5 \\ 0 & 1.5 \\ 1 & 2.5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (1, -1) = \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix}$$

$$\hat{\mathbf{Y}} = \mathbf{X}^{(5)} = \boldsymbol{\mathcal{S}}(\mathbf{X}^{(4)}) = \boldsymbol{\mathcal{S}} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.7311 & 0.2689 \\ 0.2689 & 0.7311 \\ 0.2689 & 0.7311 \\ 0.2689 & 0.7311 \end{pmatrix}$$

$$\mathcal{L} = x^{(6)} = -\frac{1}{N}\mathbf{1}^t(\mathbf{Y} \odot \log \hat{\mathbf{Y}})\,\mathbf{1} = -\frac{1}{4}(1,1,1,1) \begin{pmatrix} -0.3132 & 0 \\ 0 & -0.3132 \\ 0 & -0.3132 \\ -1.3134 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 0.5633$$

Backward: para un dato y generalización a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} = -\boldsymbol{y}^t \oslash \hat{\boldsymbol{y}}^t$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} = -\frac{1}{N}(\mathbf{Y} \oslash \hat{\mathbf{Y}})^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial \boldsymbol{x}^{(4)}} = (\hat{\boldsymbol{y}} - \boldsymbol{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{Y}}} \frac{\partial \hat{\mathbf{Y}}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\mathbf{Y}} - \mathbf{Y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} \frac{\partial \boldsymbol{x}^{(4)}}{\partial \mathbf{W}^{(3)}} = \boldsymbol{x}^{(3)} \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} \frac{\partial \boldsymbol{x}^{(4)}}{\partial \boldsymbol{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \boldsymbol{b}^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} \frac{\partial \boldsymbol{x}^{(4)}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(4)}} \mathbf{W}^{(3)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \frac{\partial \boldsymbol{x}^{(3)}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \odot \mathrm{ReLU}'(\boldsymbol{x}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \mathrm{ReLU}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(1)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\mathbf{Y}} - \mathbf{Y})^t = \begin{pmatrix} -0.0672 & 0.0672 & 0.0672 & -0.1828 \\ 0.0672 & -0.0672 & -0.0672 & 0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}\right)^t = \begin{pmatrix} -0.1828 & 0.1828 \\ -0.2890 & 0.2890 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(3)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}\right)^t = (-0.1156, 0.1156)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t}\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \begin{pmatrix} -0.1344 & 0.1344 & 0.1344 & -0.3656 \\ 0.1344 & -0.1344 & -0.1344 & 0.3656 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \mathrm{ReLU}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 0 & 0 & 0 & -0.3656 \\ 0.1344 & -0.1344 & -0.1344 & 0.3656 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0.1344 \\ 0 & -0.1344 \\ 0 & -0.1344 \\ -0.3656 & 0.3656 \end{pmatrix} = \begin{pmatrix} -0.3656 & 0.2312 \\ -0.3656 & 0.2312 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = (1,1,1,1) \begin{pmatrix} 0 & 0.1344 \\ 0 & -0.1344 \\ 0 & -0.1344 \\ -0.3656 & 0.3656 \end{pmatrix} = (-0.3656, 0.2312)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} -0.1828 & 0.1828 \\ -0.2890 & 0.2890 \end{pmatrix} = \begin{pmatrix} 1.0183 & -1.0183 \\ -0.9711 & 0.9711 \end{pmatrix}$$

$$\boldsymbol{b}^{(3)t} = \boldsymbol{b}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial b^{(3)}} = (1, -1) - 0.1(-0.1156, 0.1156) = (1.0116, -1.0116)$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} -0.3656 & 0.2312 \\ -0.3656 & 0.2312 \end{pmatrix} = \begin{pmatrix} 1.0366 & 0.9769 \\ 1.0366 & 0.9769 \end{pmatrix}$$

$$\boldsymbol{b}^{(1)t} = \boldsymbol{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = (-1, 0.5) - 0.1(-0.3656, 0.2312) = (-0.9634, 0.4769)$$

In [ ]:

```python
import numpy as np; np.set_printoptions(precision=4); import torch; from torch import nn
class NeuralNetwork(nn.Module):
    def __init__(self):
        super().__init__(); self.x2 = nn.Linear(2, 2); nn.init.ones_(self.x2.weight)
        self.x2.bias.data = torch.tensor([-1., .5], requires_grad=True)
        self.x3 = nn.ReLU(); self.x4 = nn.Linear(2, 2)
        self.x4.weight.data = torch.tensor([[1., -1.], [-1., 1]], requires_grad=True)
        self.x4.bias.data = torch.tensor([1., -1.], requires_grad=True); self.x5 = nn.Softmax(1)
    def forward(self, x):
        return self.x5(self.x4(self.x3(self.x2(x))))
model = NeuralNetwork(); # print(model)
X = torch.tensor([[0., 0.], [0., 1.], [1., 0.], [1., 1.]]); print(f'X = {str(X.detach()).replace('\n','')}')
y = torch.tensor([[1., 0.], [0., 1.], [0., 1.], [1., 0.]]); print(f'y = {str(y.detach()).replace('\n','')}')
X2 = model.x2(X); print(f'X2 = {str(X2.detach()).replace('\n','')}')
X3 = model.x3(X2); print(f'X3 = {str(X3.detach()).replace('\n','')}')
X4 = model.x4(X3); print(f'X4 = {str(X4.detach()).replace('\n','')}')
X5 = model.x5(X4); print(f'X5 = {str(X5.detach()).replace('\n','')}')
loss = nn.CrossEntropyLoss()(X4, y); print(f'loss = {loss:.4f}')
model.train(); loss.backward()
torch.optim.SGD(model.parameters(), lr=1e-1).step()
print(f'W3 = {str(model.x4.weight.detach()).replace('\n','')}')
print(f'b3 = {str(model.x4.bias.detach()).replace('\n','')}')
print(f'W1 = {str(model.x2.weight.detach()).replace('\n','')}')
print(f'b1 = {str(model.x2.bias.detach()).replace('\n','')}')
```

```
X = tensor([[0., 0.],        [0., 1.],        [1., 0.],        [1., 1.]])
y = tensor([[1., 0.],        [0., 1.],        [0., 1.],        [1., 0.]])
X2 = tensor([[-1.0000,  0.5000],        [ 0.0000,  1.5000],        [ 0.0000,  1.5000],        [ 1.0000,  2.5000]])
X3 = tensor([[0.0000, 0.5000],        [0.0000, 1.5000],        [0.0000, 1.5000],        [1.0000, 2.5000]])
X4 = tensor([[ 0.5000, -0.5000],        [-0.5000,  0.5000],        [-0.5000,  0.5000],        [-0.5000,  0.5000]])
X5 = tensor([[0.7311, 0.2689],        [0.2689, 0.7311],        [0.2689, 0.7311],        [0.2689, 0.7311]])
loss = 0.5633
W3 = tensor([[ 1.0183, -0.9711],        [-1.0183,  0.9711]])
b3 = tensor([ 1.0116, -1.0116])
W1 = tensor([[1.0366, 1.0366],        [0.9769, 0.9769]])
b1 = tensor([-0.9634,  0.4769])
```

6 Ejercicio: XOR con salida sigmoide

Modelo: capa oculta con dos ReLUs y capa de salida con sigmoide para dos clases

$$\boldsymbol{x}^{(1)} \in \{0,1\}^2$$

$$\boldsymbol{x}^{(2)} = \mathbf{W}^{(1)}\boldsymbol{x}^{(1)} + \boldsymbol{b}^{(1)} \in \mathbb{R}^2$$

$$\boldsymbol{x}^{(3)} = \mathrm{ReLU}(\boldsymbol{x}^{(2)}) \in \mathbb{R}^2$$

$$x^{(4)} = \mathbf{W}^{(3)}\boldsymbol{x}^{(3)} + b^{(3)} \in \mathbb{R}$$

$$x^{(5)} = \hat{y} = \sigma(x^{(4)}) \in [0,1]$$

$$x^{(6)} = \mathcal{L} = \mathbb{H}(y,\hat{y}) = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$$

Parámetros actuales:

$$\mathbf{W}^{(1)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \mathbf{W}^{(3)} = \begin{pmatrix} 1 & -1 \end{pmatrix} \quad b^{(3)} = 0$$

Objetivo: backprop con $\mathbf{X} = \begin{pmatrix} \boldsymbol{x}_1^t \\ \boldsymbol{x}_2^t \\ \boldsymbol{x}_3^t \\ \boldsymbol{x}_4^t \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}$ y $\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$

Solución:

Forward:

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)}\mathbf{W}^{(1)t} + \mathbf{1}\boldsymbol{b}^{(1)t} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} (0, -1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}$$

$$\mathbf{X}^{(3)} = \mathrm{ReLU}(\mathbf{X}^{(2)}) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}$$

$$\mathbf{X}^{(4)} = \mathbf{X}^{(3)}\mathbf{W}^{(3)t} + \mathbf{1}b^{(3)t} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} 0 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\hat{\boldsymbol{y}} = \mathbf{X}^{(5)} = \boldsymbol{\sigma}(\mathbf{X}^{(4)}) = \begin{pmatrix} 0.5000 \\ 0.7310 \\ 0.7310 \\ 0.7310 \end{pmatrix}$$

$$\mathcal{L} = x^{(6)} = -\frac{1}{N}(\boldsymbol{y}^t \log(\hat{\boldsymbol{y}}) + (\mathbf{1} - \boldsymbol{y})^t \log(\mathbf{1} - \hat{\boldsymbol{y}})) = -\frac{1}{4}(2\log 0.7310 + \log 0.5 + \log 0.2690) = 0.6582$$

Backward: para un dato y generalización a batch

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} = \frac{1}{N}(\hat{\boldsymbol{y}} - \boldsymbol{y})^t \oslash (\hat{\boldsymbol{y}} \odot (\mathbf{1} - \hat{\boldsymbol{y}}))^t$$

$$\frac{\partial \mathcal{L}}{\partial x^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial x^{(4)}} = \hat{y} - y$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{\partial \mathcal{L}}{\partial \hat{\boldsymbol{y}}} \frac{\partial \hat{\boldsymbol{y}}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\boldsymbol{y}} - \boldsymbol{y})^t$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial \mathbf{W}^{(3)}} = \boldsymbol{x}^{(3)} \frac{\partial \mathcal{L}}{\partial x^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \odot 1^t$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial b^{(3)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \frac{\partial x^{(4)}}{\partial \boldsymbol{x}^{(3)}} = \frac{\partial \mathcal{L}}{\partial x^{(4)}} \mathbf{W}^{(3)}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} \frac{\partial \mathbf{X}^{(4)}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t} \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \frac{\partial \boldsymbol{x}^{(3)}}{\partial \boldsymbol{x}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(3)}} \odot \mathrm{ReLU}'(\boldsymbol{x}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \frac{\partial \mathbf{X}^{(3)}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \mathrm{ReLU}'(\mathbf{X}^{(2)t})$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \mathbf{W}^{(1)}} = \boldsymbol{x}^{(1)} \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(1)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \frac{\partial \boldsymbol{x}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}^{(2)}} \odot \mathbf{1}^t$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \frac{\partial \mathbf{X}^{(2)}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t \left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} \right)^t$$

Backward: cálculos

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \frac{1}{N}(\hat{\boldsymbol{y}} - \boldsymbol{y})^t = (0.1250, -0.0673, -0.0673, 0.1828)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \mathbf{X}^{(3)t}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}\right)^t = \begin{pmatrix} 0.2310 \\ 0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial b^{(3)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}}\right)^t = 0.1732$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} = \mathbf{W}^{(3)t}\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(4)}} = \begin{pmatrix} 0.1250 & -0.0673 & -0.0673 & 0.1828 \\ -0.1250 & 0.0673 & 0.0673 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}} = \frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(3)}} \odot \mathrm{ReLU}'(\mathbf{X}^{(2)t}) = \begin{pmatrix} 0 & -0.0673 & -0.0673 & 0.1828 \\ 0 & 0 & 0 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \mathbf{X}^{(1)t}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -0.0673 & 0 \\ -0.0673 & 0 \\ 0.1828 & -0.1828 \end{pmatrix} = \begin{pmatrix} 0.1155 & -0.1828 \\ 0.1155 & -0.1828 \end{pmatrix}$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = \mathbf{1}^t\left(\frac{\partial \mathcal{L}}{\partial \mathbf{X}^{(2)}}\right)^t = (1, 1, 1, 1) \begin{pmatrix} 0 & 0 \\ -0.0673 & 0 \\ -0.0673 & 0 \\ 0.1828 & -0.1828 \end{pmatrix} = (0.0482, -0.1828)$$

Actualización de parámetros: capa a capa, respetando el formato (transpuesto) del modelo

$$\mathbf{W}^{(3)t} = \mathbf{W}^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(3)}} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.2310 \\ 0.1828 \end{pmatrix} = \begin{pmatrix} 0.9769 \\ -1.0183 \end{pmatrix}$$

$$b^{(3)t} = b^{(3)t} - \eta \frac{\partial \mathcal{L}}{\partial b^{(3)}} = 0 - 0.1 \cdot 0.1732 = -0.0173$$

$$\mathbf{W}^{(1)t} = \mathbf{W}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(1)}} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} - 0.1 \begin{pmatrix} 0.1155 & -0.1828 \\ 0.1155 & -0.1828 \end{pmatrix} = \begin{pmatrix} 0.9885 & 1.0183 \\ 0.9885 & 1.0183 \end{pmatrix}$$

$$\boldsymbol{b}^{(1)t} = \boldsymbol{b}^{(1)t} - \eta \frac{\partial \mathcal{L}}{\partial \boldsymbol{b}^{(1)}} = (0, -1) - 0.1\,(0.0482, -0.1828) = (-0.0048, -0.9817)$$

In [ ]:

```python
import numpy as np; np.set_printoptions(precision=4); import torch; from torch import nn
class NeuralNetwork(nn.Module):
    def __init__(self):
        super().__init__(); self.x2 = nn.Linear(2, 2); nn.init.ones_(self.x2.weight)
        self.x2.bias.data = torch.tensor([0., -1.], requires_grad=True)
        self.x3 = nn.ReLU(); self.x4 = nn.Linear(2, 1)
        self.x4.weight.data = torch.tensor([[1., -1.]], requires_grad=True)
        nn.init.zeros_(self.x4.bias); self.x5 = nn.Sigmoid()
    def forward(self, x):
        return self.x5(self.x4(self.x3(self.x2(x))))
model = NeuralNetwork(); # print(model)
X = torch.tensor([[0., 0.], [0., 1.], [1., 0.], [1., 1.]]); print(f'X = {str(X.detach()).replace('\n','')}')
y = torch.tensor([[0.], [1.], [1.], [0.]]); print(f'y = {str(y.detach()).replace('\n','')}')
X2 = model.x2(X); print(f'X2 = {str(X2.detach()).replace('\n','')}')
X3 = model.x3(X2); print(f'X3 = {str(X3.detach()).replace('\n','')}')
X4 = model.x4(X3); print(f'X4 = {str(X4.detach()).replace('\n','')}')
X5 = model.x5(X4); print(f'X5 = {str(X5.detach()).replace('\n','')}')
loss = nn.BCELoss()(X5, y); print(f'loss = {loss:.4f}')
model.train(); loss.backward()
torch.optim.SGD(model.parameters(), lr=1e-1).step()
print(f'W3 = {str(model.x4.weight.detach()).replace('\n','')}')
print(f'b3 = {str(model.x4.bias.detach()).replace('\n','')}')
print(f'W1 = {str(model.x2.weight.detach()).replace('\n','')}')
print(f'b1 = {str(model.x2.bias.detach()).replace('\n','')}')
```

```
X = tensor([[0., 0.],        [0., 1.],        [1., 0.],        [1., 1.]])
y = tensor([[0.],        [1.],        [1.],        [0.]])
X2 = tensor([[ 0., -1.],        [ 1.,  0.],        [ 1.,  0.],        [ 2.,  1.]])
X3 = tensor([[0., 0.],        [1., 0.],        [1., 0.],        [2., 1.]])
X4 = tensor([[0.],        [1.],        [1.],        [1.]])
X5 = tensor([[0.5000],        [0.7311],        [0.7311],        [0.7311]])
loss = 0.6582
W3 = tensor([[ 0.9769, -1.0183]])
b3 = tensor([-0.0173])
W1 = tensor([[0.9884, 0.9884],        [1.0183, 1.0183]])
b1 = tensor([-0.0048, -0.9817])
```